

Fractional SPDE with Time-independent Noise: Solvability and Exact Asymptotics*

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Abstract

In this article, we study the

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1 Introduction

In this paper we study the following fractional stochastic equation:

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (1.1) \quad \boxed{\text{E:SPDE}}$$

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where $a \in (0, 2)$, $b \in (0, 2)$, $r \geq 0$, $\nu > 0$ and $\theta > 0$. Here the noise $W = \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$ is a centered and time-independent Gaussian process, defined on a complete probability space $(\Omega, \mathcal{F}, P)$, with mean zero and covariance

$$\mathbb{E}[W(\varphi)W(\psi)] = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}},$$

where μ refers to the *spectral measure*, which is assumed to be a nonnegative and nonnegative definite tempered measure on $\mathbb{R}^d$. Let γ be the Fourier transform of μ , which is also a nonnegative and nonnegative measure on $\mathbb{R}^d$ thanks to Bochner's theorem. Throughout of the paper, we use $\mathcal{F}\varphi(\xi) = \int_{\mathbb{R}^d} \exp(-ix\xi) \varphi(x) dx$ to denote the Fourier transform of a test function φ .

In (1.1), $(-\Delta)^{a/2}$ refers to the *fractional Laplacian* of order a , ∂_t^b denotes the *Caputo fractional differential operator*

$$\partial_t^b f(t) := \begin{cases} \frac{1}{\Gamma(m-b)} \int_0^t d\tau \frac{f^m(\tau)}{(t-\tau)^{b+1-m}} & \text{if } m-1 < b < m, \\ \frac{d^m}{dt^m} f(t) & \text{if } b = m, \end{cases}$$

where m is an integer, and I_t^r refers to the *Riemann-Liouville fractional integral* of order $r > 0$

$$I_t^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s) ds, \quad \text{for } t > 0,$$

with the convention that when $r = 0$, $I_t^0 = \text{Id}$ reduces to the identity operator. The fundamental solutions to (1.1) consist of a triplet:

$$\left\{ Z_{a,b,r,\nu,d}(t, x), Z_{a,b,r,\nu,d}^*(t, x), Y_{a,b,r,\nu,d}(t, x) : t > 0, x \in \mathbb{R}^d \right\}.$$

Each member of the triplet can be expressed explicitly in terms of the *Fox H-function* which is much more complicated than the fundamental function for either the heat or wave equation; See Theorem 4.1 of [5]. Since we are interested in the constant one initial condition (and zero initial velocity when $b > 1$), Theorem 4.1 (*ibid.*) implies that the corresponding solution to the homogeneous equation (i.e. the solution when there is no driving source) is equal to the constant one. Hence, by setting $G(t, x) := Y_{a,b,r,\nu,d}(t, x)$, (1.1) can be written as the following integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds, \quad (1.2)$$

E:spdeint

where the stochastic integral is in the *Skorohod* sense; see Definition 2.1 below.

We now need to make mention of the suitable choices for the noise W . Let $k \in \{1, \dots, d\}$ and partition the d -coordinates of $x = (x_1, \dots, x_n)$ into k distinct groups of size d_i for $i \in \{1, \dots, k\}$. Note that $d_1 + \dots + d_k = d$. Denote $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$ to be the coordinates in the i^{th} partition. We consider a Gaussian Noise with correlation function given by

$$\gamma(x) = \prod_{i=1}^k |x_{(i)}|^{-\alpha_i} \quad (1.3)$$

E:corfuncti

where $\alpha_i \in (0, d_i)$ and we define $\alpha = \sum_{i=1}^k \alpha_i$. This correlation function corresponds with the spectral density given by $\mu(d\xi) = \varphi(\xi) d\xi$ with

$$\varphi(\xi) = C \prod_{i=1}^k |\xi_{(i)}|^{-(d_i - \alpha_i)}. \quad (1.4)$$

E:speconfunct

The constant C can be calculated through the relation $\mathcal{F}\gamma = \varphi$. An important fact stated in [1] is that there exists a nonnegative function K on $\mathbb{R}^d$ such that $\gamma = K * K$ where $*$ denotes the spatial convolution.

Example 1.1. When $k = 1$ above then we recover the Riesz kernel $\gamma(x) = |x|^{-\alpha}$ and its spectral density $\varphi(x) = C_{d,\alpha}|x|^{-(d-\alpha)}$. In addition, we may write $\gamma = K * K$ where $*$ denotes the convolution and $K(x) = \beta_{\alpha,d}|x|^{-(d+\alpha)/2}$. In this case we have

$$C_{d,\alpha} = \pi^{-d/2} 2^{-\alpha} \frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)} \quad \text{and} \quad \beta_{\alpha,d} = \pi^{-d/4} \frac{\Gamma((d+\alpha)/4)}{\Gamma((d-\alpha)/4)} \sqrt{\frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)}}. \quad (1.5)$$

Example 1.2. When $k = d$ above then we recover the *fractional noise*. In this case, the correlation function and the spectral measure are equal to

$$\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i} \quad \text{and} \quad \varphi(\xi) = \prod_{i=1}^d C_{1,\alpha_i} |\xi_i|^{-(1-\alpha_i)},$$

respectively, where $\alpha_i \in (0, 1)$. In this case, We may also write $\gamma = K * K$ where $K(x) = \prod_{i=1}^d \beta_{\alpha_i,1} |x_i|^{-(1+\alpha_i)/2}$ where C_{1,α_i} and $\beta_{\alpha_i,1}$ are given by (1.5). If we consider the parameterization $\alpha_i = 2 - 2H_i$ with $H_i \in (1/2, 1)$, then the corresponding noise is the fractional Brownian sheet with indices $H_1, \dots, H_d$. The stochastic heat equation with such noise was studied in Hu [9].

Remark 1.3. We may actually apply the techniques in this paper to any such noise whose correlation function and spectral density, γ and μ , satisfy

1. both μ and γ are absolutely continuous with respect to Lebesgue measure
2. for some $\alpha \in (0, d)$, γ satisfies that for all $c > 0$ and $x \in \mathbb{R}^d$ that $\gamma(cx) = c^{-\alpha} \gamma(x)$
3. there exists a nonnegative function K on $\mathbb{R}^d$ such that $\gamma = K * K$ where $*$ refers to the spatial convolution.

However, there is an issue with applying the results in this paper to the more general noises satisfying 1.3. Proving ρ , defined below, is finite is a challenge. In [3], ρ is proven to be finite in the case where φ is the Riesz kernel. Below in Lemma 2.5, we give a proof showing that ρ is finite for the noise with spectral density defined above in (1.4).

Are there other noises satisfying 1.3 other than the general fractional noise?

Next, we need to make some assumption on the parameters a, b, r of (1.1) as well as the dimension d . Under these assumptions, the fundamental function G will be nonnegative and smooth for $x \neq 0$; see Theorems 4.1 and 4.6 of [4].

Assumption 1.4. The parameters a, b, r and the dimension d must satisfy one of the following conditions:

1. $d \geq 1$, $b \in (0, 1)$, $a \in (0, 2]$, $r \geq 0$;
2. $d \geq 1$, $b = 1$, $a \in (0, 2]$, $r = 0$ or $r > 1$;
3. $1 \leq d \leq 3$, $1 < b < a \leq 2$, $r > 0$;
4. $1 \leq d \leq 3$, $1 < b = a < 2$, $r > \frac{d+3}{2} - b$.

We now introduce the following variational constant, which plays an important role in this paper:

$$\mathcal{M}_a(\Theta f, \theta) := \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, f \rangle^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\} \quad \text{and} \quad \mathcal{M}_a := \mathcal{M}_a(| \cdot |^{-\alpha}, 1);$$

see Section 2.2 for more notation and details. Our goal is to develop exact moment asymptotics to the solution $u(t, x)$. We follow the techniques laid out in [1] where the stochastic wave equation, namely, $a = b = \nu = 2$ and $r = 0$, has been exclusively studied. This paper acts as a generalization of that paper to the fraction stochastic equation. Here is the main result of this paper.

T:main

Theorem 1.5. *Under Assumption 1.4, equation (1.1) driven by the noise with spectral density given in equation (1.4) has a unique solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$, $t > 0$, and $x \in \mathbb{R}^d$ provided that*

$$\alpha < \min \left(\frac{a}{b} [2(b+r) - 1], 2a, d \right). \quad (1.6)$$

Moreover, we have the following exact moment asymptotic:

$$\begin{aligned} \lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &= \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\ &\times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned} \quad (1.7)$$

E:momentasy

where

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad \text{and} \quad t_p^\beta := (p-1)^{\beta-1} t^\beta.$$

In particular, for any $p \geq 2$ and $t > 0$ fixed, we have

$$\begin{aligned} \lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) &= p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\ &\times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned} \quad (1.8)$$

E:momentasy

and

$$\begin{aligned} \lim_{p \rightarrow \infty} p^{-\beta} \log \mathbb{E}(|u(t, x)|^p) &= t^\beta \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\ &\times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned} \quad (1.9)$$

E:momentasy

respectively.

Remark 1.6. (1) By setting

$$a = b = \nu = 2 \quad \text{and} \quad r = 0,$$

we have that $t_p = (p-1)^{1/(4-\alpha)} t$, and results in (1.7), (1.8), and (1.9) recover the corresponding results for the stochastic wave equation, namely, (1.9), (1.10), and (1.11) of [1], respectively.

(2) As for the stochastic heat equation case, we are not aware of the exact asymptotics results for SHE with time-independent noise case. However, as explained in Remark 1.9 of [1], one may apply Chen's asymptotic results in [6] for SHE with time-dependent noise, by setting $H_0 = 1$, to obtain the corresponding time-independent results. In particular, by setting

$$a = b = 2, \quad \nu = 1 \quad \text{and} \quad r = 0,$$

we have that $t_p = (p-1)^{2/(4-\alpha)}t$, and results in (1.7) and (1.8) recover the corresponding results for the stochastic heat equation, namely, (1.16) and (1.17) of [1], respectively.

In the following, we first give some preliminaries in Section 2 including the Malliavin calculus, the notation of the solution, and some variational constants. The existence and uniqueness of a $L^p(\Omega)$ solution is proven in Section 3. The exact asymptotic for the case $p = 2$ is obtained in Section 4. The upper bound and lower bound for (1.7) for general $p \geq 2$ are established in Sections 5 and 6, respectively.

2 Some preliminaries

S:Pre

2.1 Malliavin Calculus

S:Malliavin

To briefly summarize [8], we start with a nonnegative and nonnegative definite tempered measure Γ with density γ in the sense that $\Gamma(dx) = \gamma(x)dx$, and

$$\int_{\mathbb{R}^d} \Gamma(dx) (\varphi * \tilde{\varphi})(x) \geq 0 \quad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^d)$$

where $\tilde{\varphi}(x) = \varphi(-x)$. In addition we require the existence of an $r > 0$ such that

$$\int_{\mathbb{R}^d} \Gamma(dx) \frac{1}{(1 + |x|^2)^r} < \infty.$$

According to the Bochner theorem, there exists a nonnegative measure μ , often referred to as the spectral measure on $\mathbb{R}^d$ whose Fourier transform is Γ in the sense that for any $\varphi \in (\mathbb{R}^d)$

$$\int_{\mathbb{R}^d} \Gamma(dx) \varphi(x) = \int_{\mathbb{R}^d} \mu(dt) \mathcal{F}\varphi(t)$$

The functional defined on the space of test functions $\mathcal{D}(\mathbb{R}^d)$

$$C(\varphi, \psi) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi)$$

is nonnegative definite and thus we may associate it as the covariance functional of a zero-mean Gaussian processes, $W := \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$. In other words,

$$\mathbb{E}(W(\varphi)W(\psi)) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}}.$$

Let $\mathcal{H}$ be the completion of $\mathcal{D}(\mathbb{R}^d)$ with respect to $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and thus we see $\varphi \mapsto W(\varphi)$ is an isometry from the normed space $(\mathcal{D}(\mathbb{R}^d), \|\cdot\|_{\mathcal{H}})$ to $L^2(\Omega)$ where $\|f\|_{\mathcal{H}} = \langle f, f \rangle_{\mathcal{H}}$.

We denote δ the Skorohod integral with respect to W and we denote its domain $\text{Dom}(\delta)$. For any $u \in \text{Dom}(\delta)$, we write

$$\delta(u) = \int_{\mathbb{R}^d} u(x) W(dx).$$

Through construction of δ , $\delta(u)$ is a random variable in $L^2(\Omega)$. For a deeper explanation, we direct the reader to Nualart [11].

D:Sol

Definition 2.1. A random field $u = \{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$ such that $\|u(t, x)\|_2 < \infty$ for all $t \geq 0$ and $x \in \mathbb{R}^d$ is called a *mild solution* to the equation (1.1) if the random field $\{G(t-s, x-y)u(s, y) : s \in [0, t], y \in \mathbb{R}^d\}$ is Skorohod integrable for each $s \in [0, t]$ and $x \in \mathbb{R}^d$ fixed and satisfies the following integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y)u(s, y)W(\delta y) \right) ds. \quad (2.1)$$

Through the standard Picard iteration scheme, the solution can be expressed by

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)) \quad (2.2)$$

E:solwexp

where $I_k : \mathcal{H}^{\otimes k}((\mathbb{R}^d)^k) \rightarrow \mathcal{H}_k$ is the k^{th} order Skorohod integral and $\mathcal{H}_k$ is the k^{th} Wiener chaos space. The kernels $f_k(\cdot, x, t) \in \mathcal{H}^{\otimes k}((\mathbb{R}^d)^k)$ are calculated through the Piccard iteration scheme to be

$$\begin{aligned} f_n(x_1, \dots, x_n; x, t) &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t-t_n, x-x_n) \cdots G(t_2-t_1, x_2-x_1) dt_1 \cdots dt_n \\ &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t_1, x-x_n) \cdots G(t_n-t_{n-1}, x_2-x_1) dt_1 \cdots dt_n. \end{aligned} \quad (2.3)$$

E:wker

For ease of notation throughout this article, we write the above integrals as

$$\begin{aligned} f_n(x_1, \dots, x_n; x, t) &= \int_{[0, t]_{<}^n} G(t-t_n, x-x_n) \cdots G(t_2-t_1, x_2-x_1) d\vec{t} \\ &= \int_{[0, t]_{<}^n} G(t_1, x-x_n) \cdots G(t_n-t_{n-1}, x_2-x_1) d\vec{t}. \end{aligned}$$

where in general $d\vec{z} = \prod_{k=1}^d dz_k$ and the dimension will always be easily understood. We denote by $\widetilde{f}_n(\cdot, x, t)$ to be the symmetrization of $f_n(\cdot, x, t)$ in the variables $x_1, \dots, x_n$:

$$\widetilde{f}_n(\cdot; x, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} f_n(x_{\rho(1)}, \dots, x_{\rho(n)}) \quad (2.4)$$

$$= \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} G(t-t_n, x-x_{\rho(n)}) \cdots G(t_2-t_1, x_{\rho(2)}-x_{\rho(1)}) d\vec{t} \quad (2.5)$$

where Σ_n is the set of all permutations of $\{1, \dots, n\}$. In particular, for $x = 0$ we have

$$\widetilde{f}_n(\cdot; 0, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} G(t_1, x_{\rho(1)}) \cdots G(t_n-t_{n-1}, x_{\rho(n)}-x_{\rho(n-1)}) d\vec{t}. \quad (2.6)$$

E:fperm0

By setting $t_{n+1} = t$, the Fourier transform of the kernels, f_n , is given by

$$\mathcal{F}f_n(\cdot; x, t)(\xi_1, \dots, \xi_n) = e^{-i(\sum_{j=1}^n \xi_j) \cdot x} \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{k+1}-t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) dt. \quad (2.7)$$

E:fwker

Recall the notation above that $G(t, x) = Y_{a,b,r,d}(t, x)$. The following result shows an important scaling property that both $\mathcal{F}G(t, \cdot)(\xi)$ and $\left\| \widetilde{f}_n(\cdot, t, x) \right\|_{\mathcal{H}^{\otimes n}}$ have.

L:Scale

Lemma 2.2. For any $c > 0$ and $t > 0$, the two following scaling properties hold:

$$\mathcal{F}G(t, \cdot)(c\xi) = c^{-\frac{a}{b}(b+r-1)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi) \quad \text{and} \quad \mathcal{F}G(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}G(t, \cdot)(c^{b/a}\xi), \quad (2.8)$$

E:greenscale

and

$$\left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \quad (2.9)$$

E:normscale

Proof. The scaling properties in (2.8) is a direct consequence of the explicit expression of $\mathcal{F}G(t, \cdot)(\xi)$ as in (4.8) of [5]. Property (2.9) is a direct consequence of (2.8) and the scaling property of the spectral measure μ . We leave the details for the interested readers. $\square$

Recall the following standard result about the existence and uniqueness of the solution when the solution can be written as the Wiener chaos expansion as in (2.2).

Theorem 2.3. Fix any $T \in (0, \infty]$. Suppose that $f_n(\cdot, x, t) \in \mathcal{H}^{\otimes n}$ for any $t \in (0, T)$, $x \in \mathbb{R}^d$ and $n \geq 1$. Then equation (1.1) has a unique $L^2(\Omega)$ solution on $(0, T) \times \mathbb{R}^d$ if and only if the series (2.2) converges in $L^2(\Omega)$ for any $(t, x) \in (0, T) \times \mathbb{R}^d$, which is equivalent to the convergence of the series (2.10). In this case, the solution is given by (2.2) with the second moment given by

$$\mathbb{E}(u(t, x)^2) = \sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{for all } (t, x) \in (0, T) \times \mathbb{R}^d. \quad (2.10)$$

E:2mom

2.2 Some large deviation results

SS:LD

Let us first define

$$\rho := \rho_{\nu, a} = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx).$$

Recall that when $\mu(d\xi) = C_{d,\alpha} |\xi|^{-(d-\alpha)} d\xi$ which corresponds to the Riesz kernel $\gamma(x) = |x|^{-\alpha}$ for some $\alpha \in (0, d)$, Bass, Chen and Rosen [3] established that when $a \in (0, 2]$ and $\nu = 2$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log(\rho_{2,a}), \quad (2.11)$$

E:rhovavede

and

$$\rho_{2,a} = \Lambda_\alpha^{2-(\alpha/a)} \quad (2.12)$$

E:oldrhoavar

where

$$\begin{aligned} \Lambda_\alpha &:= \sup_{g \in \mathcal{F}_a} \left\{ \left(\int_{\mathbb{R}^{2d}} \frac{g^2(x_1)g^2(x_2)}{|x_1+x_2|^\alpha} d\vec{x} \right)^{1/2} - \mathcal{E}_a(g, g) \right\} \\ &= \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, |\cdot|^{-\alpha} \rangle_{L^2(\mathbb{R}^d)} - \mathcal{E}_a(g, g) \right\} \quad \text{and} \end{aligned} \quad (2.13)$$

$$\mathcal{E}_a(g, g) := (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi. \quad (2.14)$$

The following result shows the relationship between $\rho_{2,a}$ and ρ and well as the generalization of (2.11) for ρ .

L:rhovar

Lemma 2.4. For any $\nu > 0$ and $a \in (0, 2]$, we have that

$$\rho = \left(\frac{\nu}{2}\right)^{-\alpha/a} \Lambda_\alpha^{2-(\alpha/a)}. \quad (2.15) \quad \text{E:rhoLam}$$

In addition,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log(\rho). \quad (2.16) \quad \text{E:LogRho}$$

Proof. Recall that we define $\varphi(x) = C_{d,\alpha} |x|^{-(d-\alpha)}$. We begin by applying the substitution $x' = (\frac{\nu}{2})^{1/a}$ to ρ :

$$\begin{aligned} \rho &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1 + \frac{\nu}{2}|x+y|^a} \sqrt{1 + \frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx) \\ &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1 + |x+y|^a} \sqrt{1 + |y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \left(\frac{\nu}{2}\right)^{(d-\alpha)/a} \varphi(x) \left(\frac{\nu}{2}\right)^{-d/a} dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1 + |x+y|^a} \sqrt{1 + |y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \varphi(x) dx \end{aligned}$$

Now define $f^*(x) = (\frac{\nu}{2})^{\frac{-d}{2a}} f\left(\left(\frac{\nu}{2}\right)^{-1/a}x\right)$. Then with this we see that

$$\rho = \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y)f^*(y)}{\sqrt{1 + |x+y|^a} \sqrt{1 + |y|^a}} dy \right]^2 \varphi(x) dx$$

and note that

$$\int_{\mathbb{R}^d} f(x)^2 dx = \int_{\mathbb{R}^d} f^*(x)^2 dx$$

and thus

$$\left\{ f \in L^2(\mathbb{R}^d) : \|f\|_2 = 1 \right\} = \left\{ f \in L^2(\mathbb{R}^d) : \|f^*\|_2 = 1 \right\}$$

so we can say that

$$\begin{aligned} \rho &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y)f^*(y)}{\sqrt{1 + |x+y|^a} \sqrt{1 + |y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f^*\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y)f^*(y)}{\sqrt{1 + |x+y|^a} \sqrt{1 + |y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \rho_{2,a} \end{aligned}$$

and (2.15) is proven by applying (2.12).

To prove (2.16), we first notice that by applying the substitution $\xi'_{\sigma(j)} = (\frac{\nu}{2})^{1/a} \xi_{\sigma(j)}$ that

$$\int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi})$$

$$= \left(\frac{\nu}{2}\right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi})$$

and now we can directly calculate

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right)^2 \mu(d\vec{\xi}) \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \left(\frac{\nu}{2}\right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}) \right] \\ &= \log \left[\left(\frac{\nu}{2}\right)^{-\alpha/a} \right] + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}) \right] \end{aligned}$$

and because of both Theorem 1.5 [3] and (2.15) above,

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right)^2 \mu(d\vec{\xi}) \right) \\ &= \log \left(\left(\frac{\nu}{2}\right)^{-\alpha/a} \right) + \log(\rho_{2,a}) = \log(\rho). \end{aligned}$$

This completes the proof of Lemma 2.4. $\square$

Next we show that ρ is finite. We do this by showing $\rho_{2,a}$ is finite and then noting that this implies that ρ is finite by 2.4. We stated in the introduction that Bass, Chen and Rosen proved in [3] that ρ is finite for the case of the Riesz kernel. We mimic his proof but were he uses the Hardy-Littlewood-Sobolev theorem, we make use of the weak Young's inequality instead.

Lemma 2.5. *For any f, g, h with $h \geq 0$ and for φ given as in equation (1.4), we have the following:*

$$\left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{|f(x+y)g(y)|}{\sqrt{h(x+y)}\sqrt{h(y)}} dy \right]^p \varphi(x) dx \right)^{1/p} \leq C \|f\|_2 \|g\|_2 \|h^{-1}\|_{pd/\alpha}$$

where $\alpha = \sum_{i=1}^k \alpha_i$ with $k \in \{1, \dots, d\}$, $\alpha_i \in (0, d_i)$ and $\varphi(x) = \prod_{i=1}^k |x_{(i)}|^{-(d_i - \alpha_i)}$ where $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$ is the i^{th} group of $x = (x_1, \dots, x_d)$.

Proof. By observing the proof of Lemma 1.6 [3], we notice that we only need to prove equation (1.27) which says

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y) G(x) \varphi(x-y) dy dx \leq C \|F\|_{2d/(d+\alpha)} \|G\|_{2d/(d+\alpha)}. \quad (2.17)$$

where

$$F(x) = \frac{|f(x)|}{(h(x))^{p/2}} \quad \text{and} \quad G(x) = \frac{|g(x)|}{(h(x))^{p/2}}$$

If we can do this then the result follows by copying the rest of Chen's proof. To prove (2.17), we use the weak version of Young's inequality. Recall that weak Young's inequality [10] says

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} a(x)b(x-y)c(y)dx dy \right| \leq K_{p,q,r,d} \|a\|_p \|b\|_{q,w} \|c\|_r$$

where $p, q, r > 1$ and $1/p + 1/q + 1/r = 2$ and

$$\|b\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |f(x)| dx$$

where A is an arbitrary Borel set of finite measure. So to prove (2.17) we wish to be able to apply this with

$$p = r = 2d/(d + \alpha) \quad \text{and} \quad q = d / \left(d - \sum_{j=1}^d \alpha_j \right) = d/(d - \alpha)$$

and, in addition, with

$$a = F, \quad c = G \quad \text{and} \quad b = \varphi$$

Again, by following the steps of the proof from the Xia Chen paper, we note that it suffices to show that $\|\varphi\|_{q,w}$ is finite with $q = d/(d - \alpha)$. Note that $1/q' = 1 - 1/q = \alpha/d$.

First we define $A_R := A_1 \times \cdots \times A_k$ where $A_i \subset \mathbb{R}^{d_i}$ and $A_i = [-R, R]$ if $d_i = 1$ and $A_i = B_R(0)$ if $d_i > 1$ where $B_R(0)$ is the ball centered at the origin with radius R . With this we have that

$$\int_{A_i} |x_{(i)}|^{-(d_i - \alpha_i)} dx_{(i)} = \begin{cases} 2 \frac{R^{\alpha_i}}{\alpha_i} & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{R^{\alpha_i}}{\alpha_i} & , d_i > 1 \end{cases} \quad (2.18) \quad \boxed{\text{E:int}}$$

where we used polar coordinated to calculate the integral when $d_i > 1$ and $|\mathbb{S}^{d_i-1}|$ is the surface area of the sphere in $\mathbb{R}^{d_i}$. Also,

$$|A_i| = \begin{cases} 2R & , d_i = 1 \\ |\mathbb{S}^{d_i-1}| \frac{R^{d_i}}{d_i} & , d_i > 1. \end{cases} \quad (2.19) \quad \boxed{\text{E:leb}}$$

Recall that $1/q' = \alpha/d$. Then by combining (2.18) and (2.19) we see that

$$\begin{aligned} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx &= \left(\prod_{i=1}^k |A_i| \right)^{-1/q'} \prod_{i=1}^k \int_{A_i} |x_{(i)}|^{-(d_i - \alpha_i)} dx_{(i)} \\ &= R^{-\alpha} \left(\prod_{i=1}^k \begin{cases} 2 & , d_i = 1 \\ |\mathbb{S}^{d_i-1}|/d_i & , d_i > 1. \end{cases} \right) \times R^{\alpha} \left(\prod_{i=1}^k \begin{cases} 2/\alpha_i & , d_i = 1 \\ |\mathbb{S}^{d_i-1}|/\alpha_i & , d_i > 1 \end{cases} \right) \\ &= K \end{aligned}$$

where the constant K does not depend on R . Hence we that

$$\sup_{R>0} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx = K.$$

and by the symmetry of φ we can in fact say that

$$\sup_B |B|^{-1/q'} \int_B \varphi(x) dx = K$$

where B denotes the Borel sets of finite measure. Hence, $\varphi \in L_{q,w}(\mathbb{R}^d)$ with $q = d/(d - \alpha)$ which completes the proof of Lemma 2.5. $\square$

We finish this section by making a slight generalization of Λ_α . For this, we define

$$\mathcal{M}_a(\Theta f, \theta) = \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, f \rangle^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\}. \quad (2.20)$$

One notices that by making this generalization, we can apply Theorem 1.5 to the SPDE (1.1) driven by an arbitrary Gaussian noise with correlation function f as long as one can show that ρ is finite. By slightly modifying the proof of Lemma 2.3 [1], one can show that

$$\mathcal{M}_a(\Theta f, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(f, 1). \quad (2.21)$$

Finally we note that

$$\Lambda_\alpha = \mathcal{M}_a(|\cdot|^{-\alpha}, 2) = 2^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(|\cdot|^{-\alpha}, 1) \quad (2.22)$$

and by defining $\mathcal{M}_a := \mathcal{M}_a(|\cdot|^{-\alpha}, 1)$ and by applying Theorem 2.4 we see that

$$\rho = \left(\frac{\nu}{2}\right)^{-\alpha/a} \left[2^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a\right]^{2-(\alpha/a)} = \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}. \quad (2.23)$$

E:rhoMrep

3 Solvability

S:solvable

The goal for this section is to prove the following theorem:

T:Solvable

Theorem 3.1. *The equation (1.1) has a unique solution $\{u(t, x) : t > 0, x \in \mathbb{R}^d\}$ and*

$$\sup_{(t,x) \in [0,T] \times \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) < \infty \quad \text{for all } p \geq 2, T > 0. \quad (3.1)$$

We need to build up some theory before we prove the above theorem. We first introduce some notation that will be used throughout the remainder of his paper. We start by defining $L_n(\vec{y})$ to be the Laplace transform of $\widetilde{f}_n(\vec{y}, 0, \cdot)$ evaluated at 1:

$$\begin{aligned} L_n(\vec{y}) &= n! \int_0^\infty e^{-t} \widetilde{f}_n(\vec{y}; 0, t) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]^n} \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{s} dt \end{aligned}$$

with the convention that. $s_0 = y_{\sigma(0)} = 0$. Next we define,

$$\begin{aligned} H_n(t, \vec{x}) &= n! \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \widetilde{f}_n(\vec{y}; 0, t) d\vec{y} \\ &= \sum_{\sigma \in \Sigma_n} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{y} d\vec{s} \end{aligned}$$

where in the last line we used relation (2.6) for expressing $\widetilde{f}_n(\cdot; 0, t)$. Next, similarly to L_n , we define $L(x)$ and can calculate $\mathcal{F}L(\xi)$ to be

$$L(x) = \int_0^\infty e^{-t} G(t, x) dt \quad \text{and} \quad \mathcal{F}L(\xi) = \int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a} \quad (3.2) \quad \boxed{\text{E:LandFL}}$$

and through a change of variables, one sees that the Laplace transform of convolution in the time variable become a product:

$$L_n(\vec{y}) = \sum_{\sigma \in \Sigma_n} L(y_{\sigma(1)}) L(y_{\sigma(2)} - y_{\sigma(1)}) \cdots L(y_{\sigma(n)} - y_{\sigma(n-1)}) \quad (3.3)$$

and using this and (3.2), we can calculate $\mathcal{F}L_n(\vec{\xi})$:

$$\mathcal{F}L_n(\vec{\xi}) = \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a} \quad (3.4) \quad \boxed{\text{E:FLfn}}$$

L:Hneq1

Lemma 3.2. *Under the assumptions of Theorem 1.5, we have that*

$$\int_{(\mathbb{R}^d)^n} \left[\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right]^2 d\vec{x} = \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\vec{\xi})|^2 \mu(d\vec{\xi}). \quad (3.5)$$

Proof. Notice that

$$\begin{aligned} & \int_0^\infty e^{-t} H_n(t, \vec{x}) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{y} d\vec{s} dt \\ &= \int_{\mathbb{R}^{dn}} \prod_{k=1}^n K(x_k - y_k) L(\vec{y}) d\vec{y}. \end{aligned}$$

Therefore,

$$\begin{aligned} \int_{\mathbb{R}^{dn}} \left[\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right]^2 d\vec{x} &= \int_{\mathbb{R}^{dn}} \left[\int_{\mathbb{R}^{dn}} \prod_{k=1}^n K(x_k - y_k) L(\vec{y}) d\vec{y} \right]^2 d\vec{x} \\ &= \int_{\mathbb{R}^{dn}} |\mathcal{F}L_n(\vec{\xi})|^2 \prod_{k=1}^n \mu(d\xi_k) \end{aligned}$$

where the last line follows from Plancherel's theorem and the fact that $K * K = \gamma$. **consider adding the plancherel details** $\square$

L:inteq1

Lemma 3.3. *Under the assumptions of Theorem 1.5, we have that*

$$\left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n^2(t, \vec{x}) d\vec{x}. \quad (3.6)$$

Proof. Writing $\gamma(y_k - y'_k) = \int_{\mathbb{R}^d} K(y_k - x_k) K(y'_k - x_k) dx_k$, we see that

$$\left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \int_{(\mathbb{R}^d)^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \gamma(y_k - y'_k) \widetilde{f}_n(\vec{y}, 0; t) \widetilde{f}_n(\vec{y}', 0; t) d\vec{y} d\vec{y}'$$

$$\begin{aligned}
&= \int_{(\mathbb{R}^d)^n} \left[\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(y_k - x_k) \tilde{f}_n(\vec{y}, 0; t) d\vec{y} \right]^2 d(\vec{x}) \\
&= \int_{(\mathbb{R}^d)^n} \left[\frac{1}{n!} H_n(t, \vec{x}) \right]^2 d\vec{x}.
\end{aligned}$$

maybe add the placherel theorem steps in the above equalities. □

If f is a non-decreasing function on $[0, \infty)$, then the following reverse Cauchy-Schwarz inequality holds, which plays a key role in our developments.

Lemma 3.4. *If $f : [0, \infty) \rightarrow [0, \infty)$ is a non-decreasing function, then*

$$2 \int_0^\infty e^{-2t} f^2(t) dt \leq \left(\int_0^\infty e^{-t} f(t) dt \right)^2. \quad (3.7)$$

Proof. Let τ and τ' be two independent random variables with exponential distribution of mean one. Then $\tau \wedge \tau'$ has an exponential distribution with density $g(t) = 2e^{-2t}$ and

$$\left(\int_0^\infty e^{-t} f(t) dt \right)^2 = \mathbb{E}[f(\tau)] \mathbb{E}[f(\tau')] = \mathbb{E}[f(\tau) f(\tau')] = 2 \int_0^\infty e^{-2t} f^2(t) dt$$

□

Denote

$$T_n = T_n(\nu, a) := \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}). \quad (3.8)$$

By applying the substitution $\xi'_k = (\frac{\nu}{2})^{1/a} \xi_k$ and the scaling property for φ , we may rewrite T_n as

$$T_n = \left(\frac{\nu}{2} \right)^{-n\alpha/a} T_n(2, a) \quad (3.9)$$

Lemma 3.5. *Under the assumption of Theorem (1.7), we have that*

$$C_\mu = C_\mu(\nu, a) := \int_{(\mathbb{R}^d)} \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) < \infty \quad \text{and} \quad T_n \leq (n!)^2 (C_\mu)^2 \quad (3.10)$$

E:muassumpt

Proof. As in the proof of Lemma 3.6 [1], we apply a spherical substitution and get that

$$\int_{(\mathbb{R}^d)} \left(\frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi) = \left(\int_{S^{d-1}} \varphi(\gamma) d\sigma(\gamma) \right) \left(\int_0^\infty \frac{\rho^{\alpha-1}}{(1 + \rho^a)^2} d\rho \right).$$

The first integral is shown to be finite in [1]. The second integral is finite if

$$2a - \alpha + 1 > 1 \quad (3.11)$$

E:alpha2

which is already satisfied under the assumptions of Theorem 1.5. Hence the first relation in (3.10) is proven.

The second relation is proven the same was as in Lemma 3.6 [1]. Here is the proof.

$$T_n \leq n! \int_{(\mathbb{R}^d)^n} \sum_{\sigma \in \Sigma_n} \left[\prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n)$$

$$\begin{aligned}
&= (n!)^2 \int_{(\mathbb{R}^d)^n} \left[\prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \\
&\leq (n!)^2 \left[\sup_{\eta \in \mathbb{R}^d} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + \frac{\nu}{2} |\xi + \eta|^a} \right)^2 \mu(d\xi) \right]^n \\
&= (n!)^2 \left[\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) \right]^n \\
&= (n!)^2 (C_\mu)^n
\end{aligned} \tag{3.12}$$

where the last line follows by Lemma 4.1 of Balan and Song [2] or by Lemma 3.1 in [7]. $\square$

The following lemma shows that $\tilde{f}_n(\cdot, 0; t) \in \mathcal{H}^{\otimes n}$ for any $t > 0$.

Lemma 3.6. *Under the assumptions of Theorem 1.5 and for any $t > 0$.*

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^n dt \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n}{(n!)^2} \leq 2^{n[2(b+r) - \frac{b\alpha}{a}]} (C'_\mu)^n$$

Proof. We first derive the following scaling property for $\mathcal{F}\tilde{f}_n$ where $\tilde{f}_n$ is the kernel of the n^{th} Wiener integral in the chaos expansion of the solution (2.2):

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F}\tilde{f}_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n). \tag{3.13}$$

Let $c > 0$.

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \mathcal{F}f_n(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$$

so it suffices to just look at the $\mathcal{F}f(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$ terms individually.

$$\begin{aligned}
\mathcal{F}f_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) &= \int_{[0, ct]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) d\vec{t} \\
&= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot)} \left(\sum_{j=1}^k \xi_j \right) c^n d\vec{t}
\end{aligned}$$

where we applied the substitution $t'_i = t_i/c$ by applying the scaling property of $\mathcal{F}G$ we see that

$$\begin{aligned}
\int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot)} \left(\sum_{j=1}^k \xi_j \right) c^n d\vec{t} &= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{c^{b+r-1} \mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(c^{b/a} \sum_{j=1}^k \xi_j \right) c^n d\vec{t} \\
&= c^{n(b+r)} \mathcal{F}f_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n).
\end{aligned}$$

Now by using this scaling property we see that

$$\begin{aligned}
\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \int_0^\infty e^{-t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\tilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\
&= \int_0^\infty e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\tilde{f}_n(\cdot, 0; 2t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) 2dt
\end{aligned}$$

$$\begin{aligned}
&= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \widetilde{f}_n(\cdot, 0, t)(2^{b/a} \xi_1, \dots, 2^{b/a} \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\
&= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \widetilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 2^{-\frac{nb d}{a}} 2^{\frac{nb(d-\alpha)}{a}} \mu(d\vec{\xi}) dt \\
&= 2^{n(2(b+r)-b\alpha/a)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \widetilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\
&= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt
\end{aligned}$$

and now we apply Lemmas 3.3 and 3.4 as well as Equation (3.4) to get

$$\begin{aligned}
\frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt &\leq \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} \left(\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right)^2 d\vec{x} \\
&= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} |\mathcal{F} L_n(\vec{\xi})|^2 d\vec{\xi} \\
&= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} T_n
\end{aligned}$$

and this proves the first inequality. An application of Lemma 3.6 gives is the second inequality. $\square$

Proof of Theorem 3.1. First recall (3.13), which says

$$\mathcal{F} \widetilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F} \widetilde{f}_n(\cdot, 0, t)(c^{b/a} \xi_1, \dots, c^{b/a} \xi_n).$$

In order to prove the theorem we need to show that $\sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$. From the scaling property (2.9) we see that

$$\begin{aligned}
\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \int_0^\infty e^{-t} t^{[2(b+r)-\frac{b\alpha}{a}]n} dt \\
&= \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)
\end{aligned}$$

as long as we have

$$2(b+r) - \frac{b\alpha}{a} > 0 \tag{3.14} \quad \boxed{\text{E:alpha3}}$$

By using Lemma 3.6 we can deduce that

$$\begin{aligned}
\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \frac{\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\
&\leq \frac{2^{n[2(b+r)-\frac{b\alpha}{a}]} T_n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\
&\leq \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C'_\mu \right)^n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)}
\end{aligned}$$

By the same reasoning as in the proof in the wave paper (details in my notebook) we can say that there exists a constant depending on α , k_α , such that

$$\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \leq k_\alpha \frac{\left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} C'_\mu \right)^n}{(n!)^{2(b+r) - b\alpha/a}}$$

Therefore,

$$\begin{aligned} \sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 &\leq \sum_{n \geq 0} \theta^n n! k_\alpha \frac{\left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} C'_\mu \right)^n}{(n!)^{2(b+r) - b\alpha/a}} \\ &= \sum_{n \geq 0} \theta^n k_\alpha \frac{\left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} C'_\mu \right)^n}{(n!)^{2(b+r) - 1 - b\alpha/a}} \end{aligned}$$

and the last sum is finite when

$$2(b+r) - 1 - b\alpha/a > 0 \quad (3.15) \quad \boxed{\text{E:alpha4}}$$

□

4 The case $p = 2$

S:p=2

In this section, we will prove the following result.

T:p=2

Theorem 4.1. *Under the assumptions of Theorem 1.5, we have that*

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) = \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta (\theta \rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right).$$

In order to prove this theorem, we first need to prove the following lemma.

Lemma 4.2.

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) = \log \left(2^{2(b+r) - \frac{b\alpha}{a}} \right) + \log [\rho] \quad (4.1) \quad \boxed{\text{E:2ineq-a}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{2(b+r) - \frac{b\alpha}{a}} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \right] + \log [\rho] \quad (4.2) \quad \boxed{\text{E:2ineq-b}}$$

Proof. Recall the definition of T_n from above as well as (2.16) which says that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{T_n}{(n!)^2} \right] = \log[\rho].$$

We already showed in the proof of Theorem 3.1 above that

$$\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)$$

and then Lemma 3.6 above shows that

$$\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \leq \frac{2^{n \left[2(b+r) - \frac{b\alpha}{a} \right]}}{(n!)^2} T_n$$

Combining this and (2.16) we see that

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{1}{n} \log \left[\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right] \\ \leq \log \left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} \right) + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{T_n}{(n!)^2} \right) \\ = \log \left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} \right) + \log(\rho) \end{aligned}$$

which proves the upper bound for (4.1).

Now we prove the lower bound. Define for $t > 0$ and $t' > 0$

$$J_n(t, t') = \int_{\mathbb{R}^{nd}} H_n(t, x) H_n(t', x) dx$$

and recall Lemma 3.3 which says

$$\left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n(t, \vec{x})^2 d\vec{x} = \frac{1}{(n!)^2} J_n(t, t).$$

Thus $J_n(t, t)$ satisfies the scaling property

$$J_n(t, t) = t^{[2(b+r) - b\alpha/a]n} J_n(1, 1).$$

Let τ and $\tilde{\tau}$ be independent exponential random variables with mean 1. By lemma 3.2,

$$\begin{aligned} \mathbb{E}[J_n(\tau, \tau')] &= \int_0^\infty \int_0^\infty e^{-t} e^{-\tilde{t}} J_n(t, \tilde{t}) dt d\tilde{t} = \int_{\mathbb{R}^{dn}} \left[\int_0^\infty e^{-t} H_n(t, x) dt \right]^2 dx \\ &= \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \cdots \mu(d\xi_n) = T_n \end{aligned}$$

In addition,

$$J_n(t, t') \leq J_n(t, t)^{1/2} J_n(t', t')^{1/2} = t^{[2(b+r) - b\alpha/a]n/2} t'^{[2(b+r) - b\alpha/a]n/2} J_n(1, 1).$$

Combining all of this gives us

$$\begin{aligned} T_n &= \mathbb{E}[J_n(\tau, \tilde{\tau})] \leq \mathbb{E} \left[\tau^{[2(b+r) - b\alpha/a]n/2} \right] \mathbb{E} \left[\tilde{\tau}^{[2(b+r) - b\alpha/a]n/2} \right] J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r) - b\alpha/a]n}{2} \right)^2 J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r) - b\alpha/a]n}{2} \right)^2 (n!)^2 \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \end{aligned}$$

where the last line follows from Lemma 3.3. Thus so far we have shown

$$\frac{T_n}{(n!)^2} = \left(\frac{\nu}{2} \right)^{-n\alpha/a} \frac{T'_n}{(n!)^2} \leq \Gamma \left(\frac{[2(b+r) - b\alpha/a]n}{2} \right)^2 \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad (4.3) \quad \boxed{\text{E:2lowbd}}$$

Now, another application of Stirling's formula shows that

$$\Gamma\left(\frac{[2(b+r) - b\alpha/a]n}{2}\right)^2 \sim \Gamma([2(b+r) - b\alpha/a]n + 1) 2^{-[2(b+r) - b\alpha/a]n} C_n$$

where $C_n = 2^{-1}[2(b+r) - b\alpha/a]^{1/2}(2\pi n)^{1/2}$. We now infer from (4.3) that

$$c_\alpha \frac{2^{[2(b+r) - b\alpha/a]n}}{C_n} \frac{1}{(n!)^2} T_n \leq \Gamma([2(b+r) - b\alpha/a]n + 1) \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$$

where c_α is a constant depending on α . Since $\frac{1}{n} \log(C_n) \rightarrow 0$ as $n \rightarrow \infty$, we obtain the lower bound

$$\begin{aligned} \liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma([2(b+r) - b\alpha/a]n + 1) \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) \\ \geq \log \left(2^{[2(b+r) - b\alpha/a]} \right) + \log(\rho) \end{aligned}$$

Thus (4.1) is proven. Lastly, (4.2) follows from (4.1) and the fact that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{\Gamma(an + 1)}{(n!)^a} \right] = a \log(a).$$

□

With the above lemma, we can now prove Theorem 4.1.

Proof of Theorem 4.1. Assuming that there is an $L^2(\Omega)$ solution $u(t, x)$ and then applying the scaling property (2.9) gives us

$$\begin{aligned} \mathbb{E}(|u(t, x)|^2) &= \sum_{n \geq 0} \theta^n (n!) \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} \theta^n (n!) t^{(2(b+r) - b\alpha/a)n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} x_n R_n t^{(2(b+r) - b\alpha/a)n} \end{aligned}$$

where

$$R_n = (n!)^{2(b+r) - b\alpha/a} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{and} \quad x_n = \frac{\theta^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}}$$

(4.2) above says that

$$\frac{1}{n} \log(R_n) \rightarrow \log \left[\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho \right] \quad \text{as } n \rightarrow \infty.$$

Now define R to be

$$R = \left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho.$$

We want to find a β and A so that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} x_n R^n \left(t^{(2(b+r) - b\alpha/a)} \right)^n = A$$

By lemma A3 notice that

$$\lim_{t \rightarrow \infty} \left[\frac{1}{(\theta R)t^{2(b+r)-b\alpha/a}} \right]^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \log \sum_{n \geq 0} \frac{[(\theta R)t^{2(b+r)-b\alpha/a}]^n}{(n!)^{2(b+r)-(b\alpha/a)-1}} = 2(b+r) - (b\alpha/a) - 1$$

and by an easy algebraic manipulation

$$\begin{aligned} \lim_{t \rightarrow \infty} \left[\frac{1}{t^{2(b+r)-b\alpha/a}} \right]^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \log \sum_{n \geq 0} \frac{[(\theta R)t^{2(b+r)-b\alpha/a}]^n}{(n!)^{2(b+r)-(b\alpha/a)-1}} \\ = [2(b+r) - (b\alpha/a) - 1](\theta R)^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \end{aligned}$$

hence we see that

$$\beta = \frac{2(b+r) - b\alpha/a}{2(b+r) - (b\alpha/a) - 1} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1](\theta R)^{\frac{1}{2(b+r)-(b\alpha/a)-1}}$$

or by removing the fractions in the denominator

$$\beta = \frac{2a(b+r) - b\alpha}{2a(b+r) - (b\alpha) - a} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1](\theta R)^{\frac{a}{2a(b+r)-b\alpha-a}}$$

Finally, lemma A2 says that

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\ = \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r)-b\alpha-a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right) \end{aligned} \quad (4.4)$$

□

5 Upper Bound

upperbound

In this section we use the result obtained in Theorem 4.1 and the hypercontractivity property given by Theorem B.1 in [1] to prove the following theorem.

Theorem 5.1. *Under the assumptions of Theorem 1.5*

$$\limsup_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \leq \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta (\theta \rho)^{\frac{a}{2a(b+r)-b\alpha-a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right). \quad (5.1)$$

Proof. We denote u_λ for $\lambda > 0$ to be the the solution of the fractional equation with θ replaced with λ and $u = u_\theta$. Using theorem B1 from the wave paper we can say that for any $p \geq 2$ that

$$\|u(t, 0)\|_p \leq \|u_{(p-1)\theta}(t, 0)\|_2$$

and by the scaling property

$$\begin{aligned} \|u_{(p-1)\theta}(t, 0)\|_2^2 &= \sum_{n \geq 0} (n!)^n \theta^n (p-1)^n \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} (n!)^n \theta^n \left\| \tilde{f}_n \left(\cdot, 0, t(p-1)^{\frac{1}{2(b+r)-b\alpha/a}} \right) \right\|_{\mathcal{H}^{\otimes n}}^2 \end{aligned}$$

$$= \left\| u \left(t(p-1)^{\frac{1}{2(b+r)-b\alpha/a}}, 0 \right) \right\|_2^2$$

and from this we can see that

$$\|u(t, 0)\|_p \leq \left\| u \left(t(p-1)^{\frac{1}{2(b+r)-b\alpha/a}}, 0 \right) \right\|_2 =: \|u(t_p, 0)\|_2$$

and an application of Theorem 4.1 proves the result. $\square$

6 Lower Bound

The goal of this section is to prove the following theorem.

Theorem 6.1. *Under the assumptions of Theorem 1.5*

$$\liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \geq \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta (\theta\rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right).$$

We will need to build up some theory before being able to prove the above theorem. We start by defining the function $W_n(t, \phi)$ on $(0, \infty) \times L_C^2(\mu)$ by

$$W_n(t, \phi) := \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}G(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds \quad (6.1)$$

with $s_0 = 0$. We now give conditions for when W_n is well defined.

Lemma 6.2. *If the measure μ satisfies the relation in (3.10), then $W_n(t, \phi)$ is well defined and for any $d \geq 1$, $t > 0$ and $\phi \in L_C^2(\mu)$. Moreover,*

$$\int_0^\infty e^{-t} W_n(t, \phi) dt = \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi) \dots \mu(d\xi_n) \quad (6.2)$$

Proof. By Fubini's theorem, the left hand side of (6.2) is equal to

$$\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \int_0^\infty e^{-t} \int_{[0, t]^n} \prod_{k=1}^n \mathcal{F}G(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) d\vec{s} dt \mu(d\vec{\xi}),$$

which is equal to, by the change of variables $t - s_n = u$ and $s_k - s_{k-1} = u_k$ for $k = 1, \dots, n$,

$$\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \left(\int_0^\infty e^{-u} du \right) \prod_{k=1}^n \left(\int_0^\infty e^{-u_k} \mathcal{F}G(u_k, \cdot)(\vec{\xi}) du_k \right) \prod_{k=1}^n \mu(d\xi_k).$$

Now recall that $\int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = 1/(1 + \frac{\nu}{2} |\xi|^a)$. Then the above quantity is equal to

$$\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_j|^2} \prod_{k=1}^n \mu(d\xi_k).$$

The result it follows from applying the Cauchy-Schwarz inequality. In addition, by using the same arguments that lead to (3.12), we see the above quantity is finite. $\square$

P:Wnprop1

Proposition 6.3. For $f \in \mathcal{H}$, $t > 0$, and $p, q > 1$ with $p^{-1} + q^{-1} = 1$, it holds that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2}(q-1) \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right|$$

and as a consequence, the series $\left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right|$ converges provided that $\|u(t, 0)\|_p < \infty$, which is the case under Theorem 3.1.

Proof. Suppose $f \in \mathcal{H}$ and let

$$Z_f = \exp \left(W(f) - \frac{1}{2} \|f\|_{\mathcal{H}}^2 \right) = \sum_{n \geq 0} \frac{1}{n!} I_n(f^{\otimes n}).$$

If f is a distribution, then $f^{\otimes n}$ is the distribution in $\mathcal{S}'(\mathbb{R}^{nd})$ whose Fourier transform is given by: $\mathcal{F}f^{\otimes n}(\vec{\xi}) = \prod_{k=1}^n \mathcal{F}f(\xi_k)$. If f is a function, then $f^{\otimes n}$ is also a function and is defined by $f^{\otimes n}(\vec{x}) = \prod_{k=1}^n f(x_k)$. Now using Holder's inequality and the fact that $W(f) \sim N(0, \|f\|_{\mathcal{H}}^2)$, we obtain:

$$|\mathbb{E}[u(t, 0)Z_f]| \leq \|u(t, 0)\|_p \|Z_f\|_q = \|u(t, 0)\|_p \exp \left\{ \frac{1}{2}(q-1) \|f\|_{\mathcal{H}}^2 \right\}. \quad (6.3)$$

E:Zfholder1

Now by the orthogonality of the Wiener Chaos spaces,

$$|\mathbb{E}[u(t, 0)Z_f]| \leq \sum_{n \geq 0} \theta^{n/2} \frac{1}{n!} \mathbb{E}[I_n(f_n(\cdot, 0; t))I_n(f^{\otimes n})] = \sum_{n \geq 0} \theta^{n/2} \langle \tilde{f}_n(\cdot, 0; t), f^{\otimes n} \rangle_{\mathcal{H}^{\otimes n}}. \quad (6.4)$$

E:Zfholder2

By using the definition of $\tilde{f}_n(\cdot, 0; t)$, we see that

$$\begin{aligned} \langle \tilde{f}_n(\cdot, 0; t), f^{\otimes n} \rangle_{\mathcal{H}^{\otimes n}} &= \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mathcal{F}f(\xi_k) \overline{\tilde{f}_n(\cdot, 0; t)(\vec{\xi})} \prod_{k=1}^n \mu(d\xi_k) \\ &= \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]_{<}^n} d\vec{t} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mu(d\xi_k) \prod_{k=1}^n \mathcal{F}f(\xi_k) \prod_{k=1}^n \mathcal{F}G(t_{k+1} - t_k, \cdot) \left(\sum_{k=1}^n \xi_{\sigma(k)} \right) \\ &= \int_{[0, t]_{<}^n} d\vec{t} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mu(d\xi_k) \prod_{k=1}^n \mathcal{F}f(\xi_k) \prod_{k=1}^n \mathcal{F}G(t_{k+1} - t_k, \cdot) \left(\sum_{k=1}^n \xi_k \right) \\ &= W_n(t, \mathcal{F}f) \end{aligned}$$

where $t_{n+1} = t$ and the last step is due to the change of variables $s_1 = t - t_n, \dots, s_n = t - t_1$ and $\xi'_1 = \xi_n, \dots, \xi'_n = \xi_1$. The conclusion follows from the above centered equations and (6.3) and (6.4). $\square$

Our next goal is to prove the following proposition, which is where the motivation of the definitions of t_p and β come from.

P:Wnprop2

Proposition 6.4. For $p, q > 1$ and $p^{-1} + q^{-1} = 1$ we have that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|$$

Proof. The above inequality motivates us to find a function $f_\tau(x)$ and some τ such that

1.

$$(q-1) \|f_\tau\|_{\mathcal{H}}^2 = (p-1)^{-1} \|f_\tau\|_{\mathcal{H}}^2 = t_p^\beta \|f\|_{\mathcal{H}}^2$$

2.

$$W_n(t, \mathcal{F}f_\tau) = W_n(t_p^\beta, \mathcal{F}f)$$

since then we could just apply Proposition 6.3 to the function f_τ . Note above that we have not yet defined β or t_p . We guess f_τ takes the form $f_\tau(x) = \tau^V f(\tau^W x)$. We now mention, through direct calculation one can show that $\|f_\tau\|_{\mathcal{H}}^2$ and $W_n(t, \mathcal{F}f_\tau)$ can be calculated as follows:

$$\|f_\tau\|_{\mathcal{H}}^2 = \tau^{2(V-dW)+W\alpha} \|f\|_{\mathcal{H}}^2 \quad (6.5)$$

and

$$W_n(t, \mathcal{F}f_\tau) = \tau^{n[V-W((d-\alpha)+\frac{a}{b}(b+r))]} W_n(t\tau^{\frac{a}{b}W}, \mathcal{F}f). \quad (6.6)$$

This means we must have

$$\begin{cases} V - W((d-\alpha) + \frac{a}{b}(b+r)) &= 0 \\ (p-1)^{-1} \tau^{2(V-dW)+W\alpha} &= t\tau^{\frac{a}{b}W} \end{cases}.$$

We need to now choose an appropriate τ and we choose $\tau = (p-1)t$ and so the above system of equations becomes

$$\begin{cases} V - W((d-\alpha) + \frac{a}{b}(b+r)) &= 0 \\ (p-1)^{2(V-dW)+W\alpha-1} t^{2(V-dW)+W\alpha} &= (p-1)^{\frac{a}{b}W} t^{\frac{a}{b}W+1} \end{cases}.$$

Equating the exponents of either the $(p-1)$ term or the t term both lead to the following equation

$$2(V-dW) + W\alpha - 1 - \frac{a}{b}W = 0 \quad (6.7)$$

and so now we have two linear equations in W and V that we can solve, namely

$$\begin{cases} V - W((d-\alpha) + \frac{a}{b}(b+r)) &= 0 \\ 2(V-dW) + W\alpha - 1 - \frac{a}{b}W &= 0 \end{cases}.$$

Now if we define

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad (6.8)$$

then solving the system gives us

$$\begin{cases} W = \frac{b}{a}(\beta-1) \\ V = (\frac{a}{b}(b+r) - \alpha + d) \frac{b}{a}(\beta-1) \end{cases}.$$

Lastly if we define

$$t_p^\beta := (q-1)(p-1)^\beta t^\beta = (p-1)^{\beta-1} t^\beta \quad (6.9)$$

then Proposition 6.4 is proven by applying Proposition 6.3 to the function

$$f_\tau(x) = \tau^{(\frac{a}{b}(b+r)-\alpha+d)\frac{b}{a}(\beta-1)} f\left(\tau^{\frac{b}{a}(\beta-1)} x\right)$$

with $\tau = (p-1)t$. □

To remove the absolute value sign in Proposition 6.4, we want to identify all the $f \in \mathcal{H}$ for which $W_n(t, \mathcal{F}f)$ is nonnegative. In fact, if we consider the space

$$\mathcal{H}_+ = \{f \in \mathcal{H} : f \text{ is nonnegative and nonnegative definite}\}$$

then by Plancherel's theorem

$$W_n(t, \mathcal{F}f) = \int_{[0,t]^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n (f * \gamma)(x_k) \prod_{k=1}^n G(s_k - s_{k-1}, x_k - x_{k-1}) d\vec{x} d\vec{s} =: U_n(t, f)$$

with $s_0 = 0$ and $x_0 = 0$. Under Assumption 1.4, the Green's function for the fractional equation is nonnegative and hence under these conditions we get that $W_n(t, \mathcal{F}(f)) \geq 0$ for all the functions $f \in \mathcal{H}_+$. Hence we have that

$$0 \leq W_n(t, \mathcal{F}f) = U_n(t, f) < \infty \quad \text{for all } f \in \mathcal{H}_+ \text{ and for any of the four options in Assumption 1.4.}$$

where the finiteness comes from Proposition 6.4.

Now we define

$$W_n(t) := \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} W_n(t, \mathcal{F}(f)) \quad \text{and} \quad U_n(t) := \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} U_n(t, f)$$

and with this we see that

$$W_n(t) \geq U_n(t) \geq 0$$

Theorem 6.5. *If τ is an exponential random variable with mean 1, then*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2})$$

Proof. We start by letting τ be an exponential random variable with mean 1. With this,

$$\mathbb{E}(U_n(\tau)) = \int_0^\infty e^{-t} U_n(t) dt \geq \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \int_0^\infty e^{-t} U_n(t, f) dt$$

For an $f \in \mathcal{H}_+$ and $\|f\|_{\mathcal{H}} = 1$ we have that $\mathcal{F}f$ is nonnegative and nonnegative definite which further implies that $\mathcal{F}f$ is even. In addition, Lemma 6.2 gives us that

$$\int_0^\infty e^{-t} U_n(t, f) dt = \int_0^\infty e^{-t} W_n(t, \mathcal{F}f) dt = \int_{\mathbb{R}^{dn}} \prod_{k=1}^n \mathcal{F}f(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n)$$

Modifying the ideas in the proof of Theorem 1.2 [3] gives us the following identity:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_{\mathbb{R}^{dn}} \prod_{k=1}^n (\mathcal{F}f)(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n) = \log[\rho(\mathcal{F}f)]$$

where $\rho(\cdot)$ is defined on the space of nonnegative and nonnegative definite functions and is given by

$$\rho(g) := \sup_{\|h\|_{L_2}=1} \int_{\mathbb{R}^d} g(\xi) \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta) h(\eta)}{\sqrt{1 + \frac{\nu}{2} |\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2} |\eta|^a}} d\eta \right] \mu(d\xi).$$

Note that since g is nonnegative and since μ has a nonnegative density φ then we may assume h is also nonnegative for the remainder of this proof. With this being said, we see that for any $f \in \mathcal{H}_+$ and $\|f\|_H = 1$

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(t)) \geq \log \rho(\mathcal{F}f)$$

We now need to calculate

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f)$$

Consider a nonnegative function $h \in L^2(\mathbb{R}^d)$. The function $h(\cdot)/\sqrt{1 + \frac{\nu}{2}|\cdot|^a} \in L^2(\mathbb{R}^d)$ so that $g_h(x) := (2\pi)^{d/2} \mathcal{F}^{-1} \left(\frac{h(\cdot)}{\sqrt{1 + \frac{\nu}{2}|\cdot|^a}} \right) (x)$ is well defined. **I still need to verify the details that** under these conditions, $g \in W_{1,a}(\mathbb{R}^d)$ with $\|g\|_{W_{1,a}} = 1$. Here is a sketch of the details. **we no longer need this part since we do not need g to be continuous, only need to be nonnegative and nonnegative definite.**

$$\begin{aligned} \|g\|_{W_{1,a}}^2 &= \|g\|_{L^2}^2 + (2\pi)^{-d} \sum_{k=1}^d \int_{\mathbb{R}^d} \xi_k^a |\mathcal{F}g(\xi)|^2 d\xi \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} (1 + |\xi|^a) |\mathcal{F}g(\xi)|^2 d\xi \\ &= \int_{\mathbb{R}^d} h(\xi)^2 d\xi = 1 \end{aligned}$$

where above we used the fact that $\|g\|_{L^2}^2 = (2\pi)^{-d} \|\mathcal{F}g\|_{L^2}^2$.

We need from this point on.

Now, for notation let $\widetilde{h}(x) = h(-x)$. Note that

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta &= (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}g_h(\xi - \eta) \mathcal{F}g_h(-\eta) d\eta \\ &= (2\pi)^{-d} (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) \end{aligned}$$

Since $h(\cdot)$ is real valued then $\widetilde{\mathcal{F}g_h} = \overline{\mathcal{F}g_h} = \mathcal{F}\bar{g}_h$. Using this and the fact that $\mathcal{F}(fg) = (2\pi)^{-d} \mathcal{F}(f) * \mathcal{F}(g)$, we see that $(2\pi)^{-d} (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) = \mathcal{F}[|g_h|^2](\xi)$ and with this we see that

$$\begin{aligned} \rho(\mathcal{F}f) &= \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) \mathcal{F}[|g_h|^2](\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \langle f, |g_h|^2 \rangle_{\mathcal{H}}. \end{aligned}$$

With this we see that

$$\sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \sup_{\|h\|_{L^2}=1} \langle f, |g_h|^2 \rangle_{\mathcal{H}} = \sup_{\|h\|_{L^2}=1} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2}$$

where the optimal f is chosen to be $|g_h|^2 / \| |g_h|^2 \|_{\mathcal{H}}$. Now we want to show that

$$|g_h|^2 \in \mathcal{H}_+ \quad (6.10) \quad \boxed{\text{E:GH+}}$$

which would imply that the supremum can be restricted to $\mathcal{H}_+$. If this were the case then we would have that

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{\|h\|_{L^2}=1} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2}$$

and notice that

$$\begin{aligned} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2} &= \left(\int_{\mathbb{R}^d} |\mathcal{F}[|g_h|^2](\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= (2\pi)^{-d} \left(\int_{\mathbb{R}^d} |(\mathcal{F}g_h * \tilde{\mathcal{F}}g_h)(\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= \left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta \right]^2 \mu(d\xi) \right)^{1/2} \\ &= \rho^{1/2} \end{aligned}$$

which finally leads us to the equality

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2}).$$

Hence, we finish the proof by proving (6.10). First note that from the above we see that

$$\int_{\mathbb{R}^d} |\mathcal{F}|g_h|^2(\xi)|^2 \mu(d\xi) \leq \rho < \infty$$

and so $|g_h|^2 \in \mathcal{H}$. Lastly, since $\mathcal{F}|g_h|^2(\xi)$ is nonnegative for $\xi \in \mathbb{R}^d$ and for nonnegative $h \in L^2(\mathbb{R}^d)$ then $\nu(d\xi) = \mathcal{F}|g_h|^2(\xi)d\xi$ is a nonnegative tempered measure. The Bochner-Schwarz theorem then implies that $|g_h|^2$ is nonnegative and nonnegative definite. This shows that $|g_h|^2 \in \mathcal{H}_+$. $\square$

We now give scaling properties for W_n and U_n .

Lemma 6.6. *We have that*

$$W_n(t) = t^{n \frac{b}{a} (a - \frac{\alpha}{2} + \frac{a}{b} r)} W_n(1)$$

and moreover

$$U_n(t) = t^{n \frac{b}{a} (a - \frac{\alpha}{2} + \frac{a}{b} r)} U_n(1)$$

Proof. Let $\phi = \mathcal{F}f$ and $c > 0$

$$\begin{aligned} W_n(t, \phi(c^{-1} \cdot)) &= \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(c^{-1} \xi_k) \prod_{k=1}^n G(s_k - s_{k-1}, \cdot) (\xi_k + \cdots \xi_n) \prod_{i=1}^n \varphi(\xi_i) d\xi_i ds_i \\ &= c^{nd} \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n G(s_k - s_{k-1}, \cdot) (c(\xi_k + \cdots \xi_n)) \prod_{i=1}^n \varphi(c\xi_i) d\xi_i ds_i \end{aligned}$$

$$\begin{aligned}
&= c^{nd} c^{-n(d-\alpha)} c^{-n\frac{a}{b}(b+r-1)} \int_{[0,t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n G\left(c^{\frac{b}{a}}(s_k - s_{k-1}), \cdot\right) (\xi_k + \cdots \xi_n) \prod_{i=1}^n \varphi(\xi_i) d\xi_i ds_i \\
&= c^{nd} c^{-n(d-\alpha)} c^{-n\frac{a}{b}(b+r-1)} c^{-n\frac{a}{b}} \int_{[0, c^{\frac{a}{b}}t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n G(s_k - s_{k-1}, \cdot) (\xi_k + \cdots \xi_n) \prod_{i=1}^n \varphi(\xi_i) d\xi_i ds_i \\
&= c^{-n(a-\alpha+\frac{a}{b}r)} W_n\left(c^{\frac{a}{b}}t, \phi\right)
\end{aligned}$$

Now we apply this with $c^{\frac{a}{b}} = \frac{1}{t}$ to get

$$W_n\left(t, \phi(t^{\frac{b}{a}}\cdot)\right) = t^{n\frac{b}{a}(a-\alpha+\frac{a}{b}r)} W_n(1, \phi)$$

Now note that

$$\left\|\phi(t^{\frac{b}{a}}\cdot)\right\|_{L^2(\mu)}^2 = t^{-\frac{\alpha b}{a}} \|\phi\|_{L^2(\mu)}^2$$

and from this we deduce that

$$\left\|t^{\frac{\alpha b}{2a}} \phi(t^{\frac{b}{a}}\cdot)\right\|_{L^2(\mu)}^2 = \|\phi\|_{L^2(\mu)}^2 = \|f\|_{\mathcal{H}}^2$$

and in addition

$$W_n\left(t, t^{\frac{b\alpha}{2a}} \phi(t^{\frac{b}{a}}\cdot)\right) = t^{n\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} W_n(1, \phi)$$

and the result follows after taking the sup over all $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}} = 1$. $\square$

Lemma 6.7.

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \right] \geq \log \left[\frac{\rho^{1/2}}{\left(\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)\right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right]$$

Proof. Let τ be an exponential random variable with mean 1. By lemma 6.7,

$$\mathbb{E}(U_n(\tau)) = \mathbb{E}\left(\tau^{n\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}\right) U_n(1) = \Gamma\left(n\frac{b}{a}\left(a-\frac{\alpha}{2}+\frac{a}{b}r\right)+1\right) U_n(1)$$

and by Theorem 6.6 we obtain

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\Gamma\left(n\frac{b}{a}\left(a-\frac{\alpha}{2}+\frac{a}{b}r\right)+1\right) U_n(1) \right] \geq \log(\rho^{1/2}). \quad (6.11) \quad \boxed{\text{E:1b1}}$$

Recall that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{\Gamma(an+1)}{(n!)^a} \right) = a \log(a)$$

and with this we have that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{\Gamma\left(n\frac{b}{a}\left(a-\frac{\alpha}{2}+\frac{a}{b}r\right)+1\right)}{(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right] = \frac{b}{a} \left(a-\frac{\alpha}{2}+\frac{a}{b}r\right) \log \left(\frac{b}{a} \left(a-\frac{\alpha}{2}+\frac{a}{b}r\right) \right)$$

and the result is proven after multiplying and dividing $(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}$ on the inside of the log in (6.11) $\square$

Lemma 6.8. For any $k, \theta > 0$, there exists a constant $c_1 = c_1(\alpha, \mathcal{M}, k, \theta) > 0$ such that, by setting $n_t = \lfloor c_1 t \rfloor$, it holds that

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log(k^{n_t} \theta^{n_t/2} U_{n_t}(t)) \geq \left(k\sqrt{\theta}\right)^{\frac{2a}{2ab-\alpha b+2ar}} \left(\rho^{1/2}\right)^{\frac{2a}{2ab-\alpha b+2ar}} \quad (6.12) \quad \text{E:69ineq}$$

Proof. Let $\epsilon > 0$. Lemma 6.8 guarantees the existence of an $N_\epsilon > 0$ so for all $n > N_\epsilon$

$$(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \geq \exp(n(\log(R) - \epsilon)) = R^n e^{-n\epsilon}$$

where

$$R = \frac{\rho^{1/2}}{\left(\frac{b}{a}\left(a - \frac{\alpha}{2} + \frac{a}{b}r\right)\right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}}.$$

Now fix a $c > 0$ and let $n_t := \lfloor ct \rfloor$. Notice that $n_t \geq N_\epsilon$ for any $t > t_\epsilon := (N_\epsilon + 1)/c$.

Let $t > t_\epsilon$.

$$k^{n_t} \theta^{n_t/2} U_{n_t}(t) = k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t} U_{n_t}(1) \geq \frac{k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} R^{n_t} e^{-n_t \epsilon}$$

Notice that $\lfloor ct \rfloor / t \rightarrow c$ as $t \rightarrow \infty$ which means that $n_t / t \rightarrow c$ as $t \rightarrow \infty$. With this we can say

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) = c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) =: I(n_t)$$

Now,

$$\begin{aligned} I(n_t) &\geq c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left((kR\sqrt{\theta})^{n_t} \frac{t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) + c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left[\left(\frac{t}{n_t} \right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t} \frac{n_t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right] - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(\frac{n_t^{n_t}}{(n_t)!} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) - c\epsilon \end{aligned}$$

and letting ϵ tend to 0 we see that

$$I(n_t) \geq c \left[\log(k\sqrt{\theta}R) - \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right] =: h(c).$$

We can easily find the maximal c by solving $h'(c) = 0$ and doing so we get $c = (k\sqrt{\theta}R)^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}}$. After plugging in this value for c and replacing R we arrive at the following inequality

$$I(n_t) \geq (k\sqrt{\theta})^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}} (\rho_{new}^{1/2})^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}}$$

and simplifying exponents gives us (6.12). □

We now can prove Theorem 6.1.

Proof of Theorem 6.1. By proposition 6.4, for and $p, q > 0$ with $p^{-1} + q^{-1} = 1$ we have that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|.$$

We now take the supremum over all $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = k > 0$. Recall that $W_n(t, \phi) = k^n W_n(t, \phi/k)$ for $\phi \in L^2(\mu)$ and non-negativity of $W_n(t, \cdot)$ on $\mathcal{H}_+$. Let $c > 0$.

$$\begin{aligned} \|u(t, 0)\|_p &\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=k} \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &= \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \sum_{n \geq 0} k^n \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \end{aligned}$$

where $n_t = \lceil c t_p^\beta \rceil$. Now by choosing c as in Lemma 6.8 we get that

$$\begin{aligned} \liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &\geq \liminf_{n \rightarrow \infty} \left(\frac{-\frac{1}{2} t_p^\beta k^2}{t_p^\beta} + \frac{1}{t_p^\beta} \log \left[\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \right] \right) \\ &\quad - \frac{1}{2} k^2 + (k\sqrt{\theta})^{\frac{2a}{2ab-\alpha b+2ar}} (\rho^{1/2})^{\frac{2a}{2ab-\alpha b+2ar}} =: h(k) \end{aligned}$$

By maximizing h we see that h is maximized at the point $k^* = \left(\frac{2ab-\alpha b+2ar}{2aB} \right)^{\frac{2ab-\alpha b+2ar}{2a-2[2ab-\alpha b+2ar]}}$ where $B = (\sqrt{\theta} \rho^{1/2})^{\frac{2a}{2ab-\alpha b+2ar}}$. Inserting k^* into h gives us that

$$h(k^*) = B^\beta \left(\frac{2a}{2ab-\alpha b+2ar} \right)^\beta \left(\frac{2ab-\alpha b+2ar-a}{2a} \right)$$

and plugging in the value for B proves Theorem 6.1. $\square$

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